

Typed Notes — DSA / Programming Tasks

Task: Identify the list of variables and their data types for the following: 1. Bill of an Amazon order 2. A train or bus ticket 3. Current bill 4. Gas bill.

1. Amazon Order Bill

Variable	Datatype
Order number	long long int
Order date	string
Invoice number	string
Customer name	string
Seller name	string
Product name	string
Quantity	int
Unit price	double / float
Tax rate	int
Tax amount	double / float
Total amount	double / float
Payment mode	string
State code	int
Billing address	string
PAN no.	string
GST no.	string
Net amount	float
Payment transaction ID	string
Date	string
Time	string
Invoice value	float
Shopping address	string

2. Train Ticket / Bus Ticket

Variable	Datatype
Ticket no.	long / int
Date	string
Time	string
Depot	string
Route	string
From place	string
To place	string
Passengers	int
Fare	float
Total amount	float
Balance	float
Conductor ID	int
ETM ID	string
Vehicle no.	string
Transferable	string / boolean
KMs	int
EPOS no.	string
Government subsidy	float

3. Current Bill

Variable	Datatype
Bill no.	long / string
Service connection no.	string
Unique SC no.	long / int
QR code	image
Consumer name	string
Address	string
Category	int
Phase	int

3. Current Bill — Continued

Variable	Datatype
Contracted load	float
Connected load	float
Meter number	long
Multiplication factor	float
Present reading	int
Average units	int
Billed units	int
RMD	int
Energy charge	float
Fixed charge	float
Customer charge	float
ED	float
Surcharge	float
FPPCA charge	float
Government subsidy	float
Bill amount	float
Net bill	float
Due date	string
Bill date	string / Data
Bill time	string
Division	string
Section	string
District	string

4. Gas Bill

Variable	Datatype
Consumer no.	long long int
Customer name	string
Cylinder price	float / double
CGST	float / double
SGST	float / double
Final price	float / double
Invoice date	string
Order status	string

Variable	Datatype
Connection type	string

DSA Placement / Company Notes

Find the list of companies that are in drive for placements using DSA.

Top product-based companies (DSA-heavy)

Google, Amazon, Microsoft, Apple, Netflix, Meta, Facebook, LinkedIn, Twitter/X, Salesforce, Uber, Adobe, Fintech, SaaS, etc.

Major Indian service-based companies (use DSA in tests)

TCS, Infosys, Wipro, Cognizant, HCL, Tech Mahindra, Capgemini, Accenture, DXC Technology, CGI, Mindtree, L&T Infotech, Mphasis, Virtusa, Hexaware, NTT Data, Zensar, LTI, etc.

Popular Indian product / startup companies using DSA rounds

Flipkart, Ola, Swiggy, Zomato, Paytm, Razorpay, Zoho, Freshworks, InMobi, RedBus, BookMyShow, Policy Bazaar, Naukri.com, MakeMyTrip.

GATE Syllabus for DSA

Programming and Data Structures

- Programming basics: Programming in C, variables, control flow, functions, recursion.
- Linear data structures: Arrays, strings, stacks, queues, linked lists.
- Trees: General trees, binary trees, binary search trees, basic operations.
- Heaps: Binary heaps, priority queues, basic heap operations.
- Graphs: Representation, basic graph theory, concepts.
- Hashing: Hash tables, hash functions, collision handling.

Algorithms

- Searching: Linear search and binary search.
- Sorting — basic: Bubble sort, selection sort and insertion sort.
- Sorting — advanced: Merge sort, quick sort, heap sort; divide-and-conquer idea.
- Complexity: Asymptotic worst-case time and space complexity; Big-O, Omega and Theta.
- Algorithm design techniques: Greedy, dynamic programming, divide-and-conquer.
- Graph algorithms.

Task — Structure Size and Memory Alignment

Task: Create a structure variable of a student. Estimate the size and structure manually, then write a C program to find the size of the structure using the sizeof operator. Observe both outputs. Are they the same or different? Justify your answer.

Structure used:

```
#include <stdio.h>

struct student
{
    int id;
    char grade;
    float marks;
};

int main()
{
    struct student s;
    printf("size of structure student = %zu bytes\n", sizeof(s));
    return 0;
}
```

Manual size calculation

Data member	Data type	Size (bytes)
id	int	4
grade	char	1
marks	float	4
Total		9 bytes (manual calculation)

The handwritten notes state that the actual structure size can be 12 bytes on the mentioned compiler/system, rather than the manually added 9 bytes.

Why are they different? The compiler may insert padding bytes between members so that the data is aligned in memory for efficient access.

Example memory layout from the notes:

Member	Size	Offset
id	4 bytes	0-3
grade	1 byte	4
padding	3 bytes	5-7
marks	4 bytes	8-11

Therefore, the total becomes 12 bytes instead of 9 bytes. The exact size may vary depending on the compiler and system architecture.

Task 4 — Amazon Website Functions

Homework: Open an Amazon website and find out 10 different functions.

Function / variable	Datatype
Product ID	int
Product name	string
Product price	float
Quantity	int
Customer name	string
Customer email	string
Cart items	array
Shipping address	object
Payment method	string
Order status	string

Functions noted on the page:

Add to cart • Buy now • Wishlist • Your Orders • Account • Reviews • Filters • Cancel order

Question — Return Values in C

Question: Search: return 0, 1 — what the main returns to whom?

return 0 → indicates successful program execution and returns control to the operating system.

return 1 → indicates unsuccessful execution (or an error) and returns control to the operating system.

main() → returns a value to the operating system.

Task 5 — Time Complexity Using Frequency Count

Compute the time complexity using the frequency count for the following sequences.

1. For loop:

```
for (i = 1; i <= 10; i = i + 2)
{
    i = i % 10;
    printf("%d", i);
}
```

Frequency count shown in the notes: initialization ≈ 1 , condition checks ≈ 6 , increment/update ≈ 5 , body operations counted separately. The written total is approximately 22.

2. While loop:

```
i = 1;
while (i <= 50)
{
    printf("hello");
}
```

The handwritten calculation marks the shown condition/body count as 1, but the code as written does not update *i*, so literally it would not terminate. The note appears to be focusing on operation counting rather than execution correctness.

3. For loop:

```
for (i = 1; i <= 20; i++)
{
    /* body */
}
```

The handwritten frequency count shown is 21 for the loop-control checks/iterations.

Task 5 — Time Complexity (Continued)

4. For loop with $i = i + 2$:

```
for (i = 1; i <= n; i = i + 2)
{
    printf("hello");
    printf("world");
    printf("i = %d", i);
}
```

The handwritten note gives the count as approximately $5n/2 + 2$ (based on its frequency-count approach).

Important: The dominant asymptotic time complexity of this loop is $O(n)$, because the loop advances by a constant amount (2) each iteration.